Index calculus on the Curve25519-family miniatures: coverage of what has been tried (2026-11-08)

Class. A coverage report: a synthesis of the lab's measured studies, designs and screens on this family, with a ranked list of untried or under-tried directions and the two single-target studies this session ran from that list. Nothing here is a leaderboard or `benchmarks/README.md` row. Every number is copied from the named file (measured) or marked derived; every result sentence is an unverified assertion [pcc:unverified]. Written in the worktree `claude/ic-coverage-20261108`; no push, no PR. Validator errata of 2026-11-08 are applied (section 5).

Band readings. The ecbench band's default now prices the table oracles' counted inserts (owner ruling 2026-11-08, landing in PR #3726); the band this report was first written under, with inserts unpriced, is called "the pre-2026-11-08 reading" below (formerly "the committed band"), and both readings are shown side by side wherever both were computed [pcc:unverified].

Curves. $y^2 = x^3 + A x^2 + x$ over F_p, $p \equiv 5 \pmod 8$, $\#E = 8r$ cyclic; k = 20: p = 1,048,573, A = 926, r = 131,203, sqrt(r) = 362.2; sizes k = 16..32 exist (`curve25519_family family`). **Goal of the programme.** S(IC) ≤ S(matched rho-neg) across the AGENTS.md §5a band (w = 0, 0.1, 1); every verified measured improvement over the current IC baseline counts [pcc:unverified].

0. Bottom line

- **The generic track is at its collision floor.** Every table or walk design with F unknown base logarithms needs F + 1 collisions and costs at least $\Gamma(F + 3/2)/(\Gamma(F + 1)\Gamma(3/2))$ times the rho model: 1.5x (F = 1), 1.875x (F = 2), 2.19x (F = 3) (`curve25519_rho_parity_bound_20261104`, derived, Monte Carlo within 1 %). Two comparisons with the floors hold: the F = 2 walk measures 1.8x to 1.9x rho against the 1.875x floor ratio (the only floor comparison in the cited file), and the twcj walk runs 3 % above its own second-collision floor 1.329 sqrt(r) [pcc:unverified]. The other ladder rows are distances to measured rho-neg and are not comparisons with the floor ratios: the 3.6x box is C = 3 (round 16) and the 3.2x row a C = 2 box (round 25, a BSGS-boundary arm), no cited file maps either to F = 3 (3.2x to 3.6x is 1.46x to 1.65x above the F = 3 ratio 2.19x); the F = 1 distances (1.34x to 1.40x at k = 20 to 32) sit below the 1.5x floor ratio because the reference is the measured rho-neg, which runs 9 % to 13 % above its model 0.886, and not the model rho; the `f1_walk` README calls its 1.496x agreement a coincidence from which no inference is drawn [pcc:unverified]. The best arm, the F = 1 target-in-walk walk with chained jumps (`twcj`), is 1.36x to 1.40x at k = 20 and 1.37x at k = 32, with its walk 3 % above the 1.329 sqrt(r) second-collision floor [pcc:unverified]. It is one unknown away from rho (P1 := G makes it rho with a stored walk) and is not decomposition-based index calculus; the owner ruled the C = 2 and walk arms count under the autolab decision rule, and the F = 1 arms are reported as boundary results outside the ratchet.
- **The algebraic track is $1.8 \cdot 10^4$x behind at w = 0.1 and scales as r^1.03** against rho's r^0.5 (atlas L10, T13); every coordinate, model, symmetry and base family has been measured or closed by a degree argument, and the levers that worked were selection (frame, seed, t, G; 16x in two days, per the atlas before wave 4, after which the track stalled 42x above its own brute-force grid) and F4 trace replay (0.4742 [0.47418, 0.47425] end to end, 2.1x; 0.2837 [0.2413, 0.3335] with t = 4) [pcc:unverified].
- **Structure levers are exhausted on this family**: Aut = ±1, End(E) = Z[π] with least non-scalar degree 772,948, one rational 2-torsion point whose x-action maps ⟨G⟩ off itself, no 3-torsion, no Frobenius, twist in a different group (`curve25519_structure_vs_rho_20261107`, `ideas/STRUCTURE-R4.md`) [pcc:unverified].
- **What remains untried with existing tools is small-constant work on the F = 1 arm and accounting rulings** what might change the picture needs engine work (an oracle finding relations below about 14 native operations per trial at w = 0.1 without forming R ± P_i, which `curve25519_algebraic_oracle_search_20261107` found no design for) [pcc:unverified]. This session ran the two cheapest untried knob tests; the first (fewer jumps) was falsified, the second (scaling to k = 28, 32) confirmed the flat 1.37x to 1.40x distance, with 6 of 7 frozen predictions held (P4b failed) [pcc:unverified]. The accounting question the band raised is now ruled (2026-11-08): inserts are priced by default, and rho-neg is ahead at every weight under that reading only [pcc:unverified].

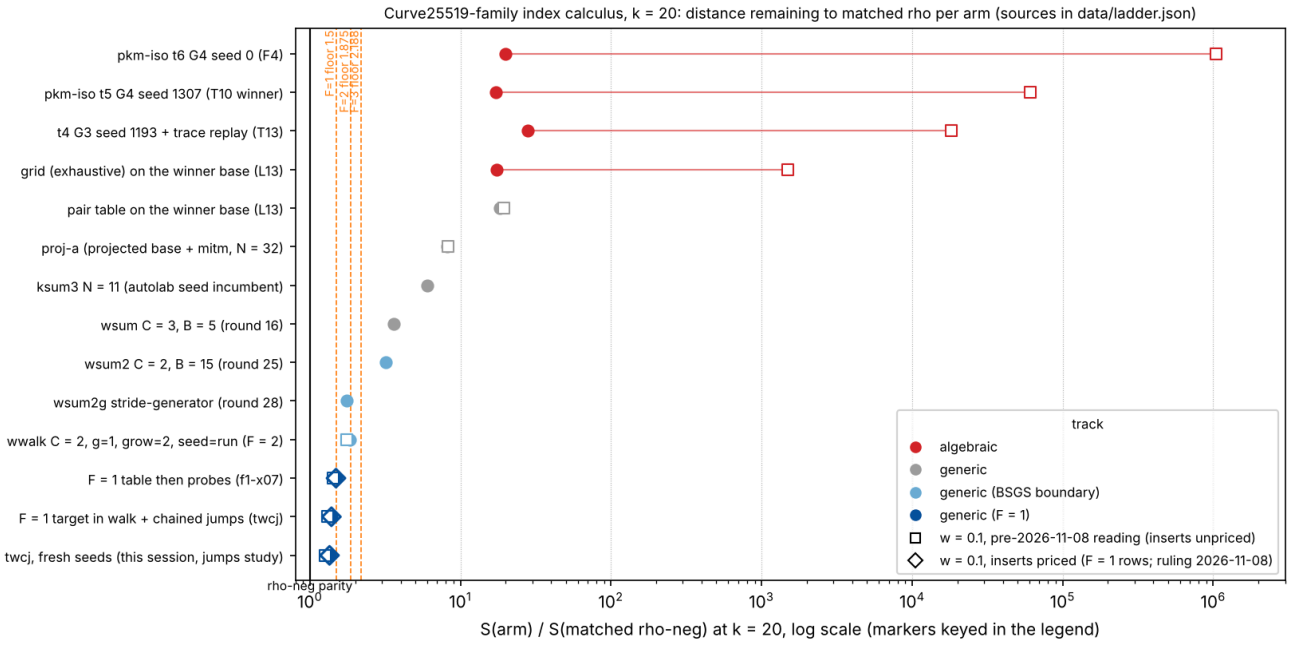


Figure 1: Distance ladder at k = 20

1. Sources read, and what could not be reached

Read in this order: `research/curve25519_ic_atlas_20261104/REPORT.md` (sections 1 to 11), `research/curve25519_autolab_20261104/README.md` and `ideas/{PROPOSALS,PROPOSALS-R3,STRUCTURE-R4,SYNTHESIS,OBSTRUCTION}.md`, `r4-i2/README.md`, `families.json`; `research/curve25519_rho_parity_bound_20261104/{README,ALGEBRAIC_SCREEN}.md`; `research/curve25519_structure_vs_rho_20261107/README.md`; `research/curve25519_algebraic_oracle_search_20261107/README.md`; `research/curve25519_f1_walk_20261107/README.md`; `research/curve25519_f1_targetwalk_20261108/README.md` and `PROTOCOL.md`; `research/curve25519_autolab_parity_20261105/` (full-size p = 2^255 - 19 endpoint, admission limits at u64, Curve25519Wide; added after validation: it contains no new scientific trial and refutes none of the claims below); `src/cryptanalysis/ecbench/methods.rs` (IC_PIPELINE_PARAMS help), `src/cryptanalysis/ic_framework/plugins.rs` (wwalk options), `src/cryptanalysis/ic_boundary.rs` (`TargetWalk`), and `ecbench methods`.

Caveat on the oracle-search study: its predictions were not preregistered, so a pre-data freeze cannot be checked for them [pcc:unverified].

Scan behind the "not found" cells (added with validator erratum E7): all 1,915 `spec*.json` files under `crypto/research` smaller than 400 KB, scanned on 2026-11-08 for the family signature (`prime_explicit` curve, p ≡ 5 mod 8, #E = 8r with r prime): 282 specs match and contain exactly two method ids, `ic.pipeline` (default linalg `incremental-gauss` throughout) and `rho.negation`. Sessions outside `spec*.json` names, the sibling repositories, the lead atlas and research-kb were not checked [pcc:unverified].

Not reachable from this machine, so no novelty claim is made anywhere below: the sibling lead atlas (`/home/user/crypto-autoresearcher/knowledge/leads/ecdlp/` does not exist on this machine; `tools/lead_atlas.py` is in the sibling checkout), the `research-kb` MCP server (not among the session's MCP servers) and its CLI fallback (`scripts/research-kb` needs the sibling checkout's executable and catalog, neither present), and `ecdb` (no `ECDB_TOKEN`). Conductor is not installed; no task was attached. Disk was at 97 % with 1.5 GB free, so the worktree is a sparse checkout and no binary was built; the measuring binary is `/home/user/ctarget-f1/release/ecbench` (SHA-256 `d23fab99…`), which already carries PR #3700's `band --price-table-work` opt-in.

2. Coverage table

Status: **done** (measured; headline number and file), **designed** (protocol or screen only, no run), **not found in specs** (no `spec*.json` under `crypto/research` with the family signature uses it; scan in section 1; not a novelty claim), **n/a** (cannot run on this family by the method's own `applies`, or does not exist on it). All values are measured unless marked derived; the distance is S(arm)/S(rho-neg) at w = 0 unless stated. Each row's number is an unverified assertion [pcc:unverified].

2a. Factor-base families (ecbench prime-curve bases and engine forms)

technique	status on this family	headline, file

`prime-projected-abscissa` (smallest x, projected by 8) + `mitm` = `proj-a`	done (baseline)	S 12.05 (T5), 12.63 (T12), 8.2x rho-neg; atlas §6
`prime-abscissa` unprojected, blind / T_0-keyed / projected	done	projection ÷ T_0 3.3 to 4.0, T_0 ÷ blind 1.94 to 2.02 yield; `curve25519_t0_symmetry_20261102` via atlas §2
`prime-interval:t,lo` (x-interval `bits`, lo = 0)	done	per relation $9.71 \cdot 10^8$ (t = 4), $9.49 \cdot 10^9$ (t = 5) mul-adds; atlas §4
`prime-interval` selected frame `bits@L` (argmax J)	done, validated	0.437 [0.300, 0.637] vs lo = 0 end to end (T11); 4.7 to 6.5x per relation; atlas §4
`prime-isogeny:t,seed` (PKM isogeny tower)	done, validated (T4 to T13)	best F4 arm: t = 4, G = 3, seed 1193 + trace, $1.83 \cdot 10^4$x rho at w = 0.1, 27.9x at w = 0 (T13); atlas §6, §11.6
`prime-kummer:t` (2-power Kummer tower)	run at t = 2 (smoke only); t > 2 n/a (p ≡ 5 mod 8: t ≤ 2)	`curve25519_ic_ratchet_20261102/sessions/pkm-smoke-k20` (`prime-kummer:t=2` with `pkm-semaev` m = 2 and m = 3 at k = 20); atlas §1, §4
`glv-orbit`	n/a (j ≠ 0, 1728, Aut = ±1)	atlas §1; structure_vs_rho C8
`cm-multiplier`	n/a (End = Z[π], least non-scalar degree 772,948 = 2134 sqrt(r))	structure_vs_rho C6b
`explicit-x`, `known-log`	instrumentation only	atlas §4
Edwards-y interval (`Membership::EdwardsBits`)	done	equal to `bits` except a t = 5, G = 0 anomaly (0.6386x) useless at the best cells (T6); atlas §2
twisted-Edwards-x, Jacobi quartic, DIK, Tate normal, Kohel μ_4, Kummer line	done by census (frames)	all Möbius-translation frames of x, one draw of the frame distribution (P1, P2 of `curve25519_frame_search_20261103`); atlas §2
T_0-invariant `t0-bits@L` (w = u + 1/u)	done, closed	0.93 to 0.95x best bits frame per relation at t = 4, 5, 1.25x at t = 6 (T7); atlas §4
2-descent `square-bits@L`	done, closed	degree 4 ⟶ 6; 1.28x / 1.71x dearer (T2); atlas §4
y-coordinate or degree-3 coordinates	closed by argument (degree)	atlas §2
Hessian, twisted Hessian, Jacobi intersection, Huff	n/a (no 3-torsion; E[2] = Z/2)	atlas §1
digit-set frames `digits@LO:…` (engine `Membership::Digits`)	done in the engine (L14/L15, validated); **not found in specs** (no ecbench base spells it; `explicit-x` could list the points; the only family mention of `explicit-x` is T11's unit tests U1 to U4, not a session)	coverage 2.1 to 2.2x the best translation frame; F4 per relation 0.48x / 0.44x; still 10.5x behind the isogeny winner at matched (t, G); atlas §11.4
odd Kummer base x ∈ g·μ_d, d | (p − 1)/4	designed (L11); decision cost screened	counted decision Ω(d), $4.3 \cdot 10^5$x at w = 0.1 for d = 1022 (oracle search D2); atlas §11.5
E[8]-closed base (one representative per orbit)	screened (counted Python)	same per-unknown cost as E[r]-only (D3, by construction); oracle search
two towers with complementary cosets	considered, not run	bounded by 1/0.75 ≈ 1.33x, below session noise; atlas §9
full-coverage m = 2 base (F ≈ 256)	screened	the "root-find per base point" is a group addition (2,000/2,000), so it is BSGS with an unknown base: 251x at w = 0.1 (D1); oracle search
isogenous curve E′ = E/⟨T_0⟩ (Z/2 × Z/4) as the working curve	screened (R4-I2 arm B)	quotient dimension divides by 2^(m−1) not 4^(m−1); D equal to arm A; Stage 1 not recommended; `autolab/r4-i2`

2b. Oracles and solvers

technique	status	headline, file
`subtract` (plain subtract-and-test)	done	dominated by ≥ 10x in autolab rounds 1 to 8; disabled after round 9; `families.json`
`mitm:m=2` negation-folded pair table	done (baseline, `mitm2`)	proj-a above; autolab incumbents at k = 28, 32 before round 4/5
`mitm:m=3`	done	worse than the incumbent at every N (round 1); disabled

`mitm-prime-batch`, `mitm-prime-blocked` (batched / blocked inversions)	done	equal S to `mitm2` (round 5): an accounting change, not an improvement; `PROPOSALS-R3` §8
`rootfind-mitm:m=2`, `rootfind-subtract:m=2`, `rootfind-pairs:m=3`	done	11.25 vs 8.70 (k = 20, round 3); pairs m = 3 70x and 320x the incumbent (rounds 3); disabled after round 9
`ksum:m=3,4,5` (signed m-sum table)	done, ratcheted	k = 24: `ksum4-n12` 0.917 [0.879, 0.957] (round 2); k = 28 0.654 (round 4); k = 32 0.352 (round 5)
`ksum … free=1` (table self-collisions as rows)	done	nulls (rounds 13 to 15): colliding 4-sums share summands, rows have low rank
`wsum:coef=B,free=1` (coefficient box, C = 3)	done, ratcheted	k = 20 0.605 [0.575, 0.637] (round 16), distance 3.6x; rounds 17 to 21
`wsum` with C = 2 (`wsum2`, BSGS boundary)	done, ratcheted; owner ruled it counts	k = 20 0.895 (round 25), distance 3.2x; B ≈ 0.47 r^(1/4), S within 3 % of 2 sqrt(3r/2) (round 26)
`targets=stride`, `stride-generator`, `stride-renew` probes	done	stride-generator ratcheted k = 20, 24, 28 (rounds 27, 28; k = 20 0.515 [0.474, 0.559], distance 1.76x fixed-table); fixed and renewed strides lose (rounds 29, 30: hits cluster on short seed relations)
`wwalk` (walk-built C = 2 table, target-free rows)	done, ratcheted	round 31; `g=1,grow=2` round 33; `seed=run` re-baseline 1.94x / 1.81x / 1.82x at k = 24 / 28 / 32 (R4-I1 correction of the fixed-draw 1.43x to 1.55x)
`wwalk … interleave=1` (growth and probes in one map)	done, falsified in a unit trend	S 2.10 to 2.15 vs 1.77 static (PROPOSALS-R3 outcomes); not merged
`wwalk` x and jump grids on F = 2	done	S flat in x (rounds 32, 35); 8 / 32 jumps nulls, tried against the fixed-table incumbent only (round 34; the autolab README states these are not evidence that the challengers would fail against the expected cost, as they were not re-run under `seed=run`)
F = 1 `wwalk` (one-abscissa base, `g=1,grow=2`)	done	0.809 [0.776, 0.842] of F = 2 at k = 20; 1.50x rho-neg at w = 0; `curve25519_f1_walk_20261107`
F = 1 target in walk (`tw=1`) and chained jumps (`cj=1`)	done	twcj 0.909 [0.885, 0.934] of F = 1 at k = 20; 1.40x rho-neg; `curve25519_f1_targetwalk_20261108`
F = 1 target-in-walk jump count 2 / 4 / 8 / 32	**done this session**	4 jumps 1.22x / 1.27x the 16-jump arm; 2 jumps 3.3x / 3.5x; 8 jumps 1.008 [0.967, 1.051] at k = 20 (tie) and 1.050 [1.005, 1.097] at k = 24 (excludes 1 by 0.005); 32 jumps 1.051 [0.996, 1.110] and 1.014 [0.978, 1.051] (ties); 16 is not beaten by any tested count; `curve25519_tw_jumps_20261108`
F = 1 target-in-walk at k = 28, 32 (and 32 / 64 jumps there)	**done this session**	1.395x [1.345, 1.446] and 1.366x [1.267, 1.473] rho-neg (k = 28, 32); 32 / 64 jumps nulls; 6 of 7 predictions held; `curve25519_tw_scaling_20261108`
`pkm-semaev:m=2,engine=f4` on the isogeny base	done, validated	T12 winner 0.0578 [0.047, 0.072] vs seed 0 at w ≥ 0.1; atlas §6
`pkm-semaev:m=3`	done, closed	2,300 to 6,600x costlier per relation than selected m = 2 (T3); atlas §3
F4 trace replay (learned refutation traces)	done, validated	0.4742 [0.47418, 0.47425] end to end (L12); T13 0.2837 [0.2413, 0.3335] vs the T12 winner at w ≥ 0.1
`pkm-semaev … engine=exhaustive` (grid)	done, validated	grid ÷ F4 0.0238 at w = 0.1 (L13); grid still 2.3x / 194x / 1,181x behind proj-a
`mitm-fiber:fiber=closed|lifts|blind`, torsion-shifted targets	done	lifts add 1.00 to 1.12x after selection (T1, failed 1.10x rule in 19 of 20 cells); atlas §3.2, §5
`descent-algebraic` (+ f4-f2, buchberger-f2, sat-cdcl, exhaustive solvers)	n/a	binary-field (F_2) solvers; help text in `methods.rs`
`gaudry-x-bin`, `pdp3-koblitz`, `mitm-frobenius*`	n/a	binary / Koblitz curves only

the 2-torsion quotient of #3408 carried to the family (R4-I2 arm A, Singular / msolve)	done (Stage 0), not passed	D drops by one at m = 3, 4 where cells finish; the m = 2 line fails; k = 20 base cells hit the cap; `autolab/r4-i2`
eliminant Φ with batched multipoint evaluation, LLL on the S_3 monomials, norm-resultant on g·μ_d	screened (counted Python)	821x, no recovery (0 of 40), $4.3 \cdot 10^5$x at w = 0.1 (D4, D7, D2); oracle search
`ic.residual_batch`, `rho.multitarget_pooled` (Kuhn–Struik batches), `ic.cm_folded_*`	not found in specs (applicability not re-checked against `applies`)	single-target scope for the first two; `ecbench methods`
`ic.large_prime`, `ic.trace_line_2lp` (BinaryOnly), `ic.shared_rank` (KoblitzOnly), `ic.k4_gaudry` (fp4_translation curves only)	n/a by `applies`	`methods.rs` help text and `applies`
`ic.prime_orbit_relations_v1`, `ic.aps_v1/v2`, `ic.tensor_contraction_v1` and its cofactor variant, `ic.known_log_ksum_v1` (PrimeOnly)	applicable, not found in specs	the earlier reason "need an orbit, a CM endomorphism or an extension field" does not apply to `ic.tensor_contraction` (a toy exponential method, about r point additions per relation); `methods.rs`

Each row's number or status in section 2b is an unverified assertion [pcc:unverified].

2c. Relation-collection strategies, linear algebra, structure, accounting

technique	status	headline, file
target stream: `walk`, `random`, strides	done (walk, strides)	rounds 12, 27 to 30 (stride tests)
`targets=walk-prime-fast`	not found in specs; relabel by help text only (native Montgomery additions; no autolab file mentions it)	`methods.rs` help text
probe-probe collisions (store probes)	not run: the rho guard (`is_guarded`) refuses it as rho with extra steps	PROPOSALS-R3 §8
multi-target amortisation (P4; `ecbench amortize`)	designed, not run; a reporting convention	atlas §9; PROPOSALS-R3 §5
seed selection by J, J/columns, joint (t, G, seed)	done, validated	T4/T5 0.641, T9 0.5605 [0.421, 0.747], T10/T12 0.0578 at w ≥ 0.1
frame selection by J (log-free)	done, validated	coverage 2.5 to 2.8x (t = 5); atlas §2
coset-pure filtering	closed by count	atlas §5
`linalg`: `incremental-gauss` (default)	done	every pipeline arm
`linalg`: `incremental-gauss-full-rank(-checked)`, `echelon`, `structured-gauss`, `sparse-cado`, `wiedemann`, `lanczos`	not found in specs; not expected to matter (F ≲ 32 columns; LA charged N^2 is negligible against the walk)	`methods.rs`; SYNTHESIS Table 2 (LA binds only at N ≈ sqrt(r/2) for m = 2)
automorphisms, Frobenius, T_0 keys, E[8] keys, twist, 2-isogenous E′, endomorphism-orbit folds, GLV, x-only ladders, DPs, tag tracing, special prime, pairings	done (exact checks and counted Monte Carlo, 7 of 8 frozen checks passed)	no lever below rho-neg across the band; folds below 1 only at w = 0 and ≥ 16.7x at w = 0.1; `curve25519_structure_vs_rho_20261107` §3
walk-table draw resampling (`seed=run`)	done (accounting correction R4-I1)	15 to 22 % of S; fixed-draw figures retired
C = 1 boundary (one-seed `wwalk`)	done	1.40 / 1.60 / 1.74 / 1.69 of rho at k = 20 / 24 / 28 / 32; `autolab/c1`
table-insert pricing in the band (PR #3700 opt-in; default after the owner ruling of 2026-11-08, PR #3726)	ruled: inserts priced by default (`--no-price-table-work` is the earlier reading)	pre-2026-11-08 reading: twcj/rho 0.80x at w = 1 is an artifact; inserts priced: 1.32x to 1.41x at every weight; `f1_targetwalk` §5, `tw_scaling` §2
rho-neg reference set-up (0.376 sqrt(r) at k = 20) and search 9 to 13 % above 0.886	observed, not acted on (a rho-implementation matter inherited by matched rho)	`structure_vs_rho` §6; `tw_scaling` §4
Weil restriction, isogeny-class transport to j = 0 / 1728, Bernstein–Lange precomputation	closed by argument	atlas §9; SYNTHESIS §2, §3

Each row's number or status in section 2c is an unverified assertion [pcc:unverified].

3. Ranked untried or under-tried directions

Ranking by expected information per hour with existing tools. Each numeric sentence below is an unverified assertion [pcc:unverified] (tagged per sentence where a prediction or derived number appears). For each: mechanism, the cheapest

discriminating single-target test, the preregistered prediction, and why it might beat the current best (twcj at 1.40x rho-neg at w = 0) or the generic floor $\Gamma(F + 3/2)/(\Gamma(F + 1)\Gamma(3/2))$ [pcc:unverified].

1. **Jump count of the target-in-walk arm** (ran this session as `curve25519_tw_jumps_20261108`). Mechanism: jump indices are drawn independently of the point, so the set-up (17 % of the k = 20 cost, about 3.5 GAE per jump) looked removable without a walk penalty. Test: `jumps=2,4,8,32` vs 16 on `twcj`, 8 single-target sessions × 64 rounds at k = 20 and 24. Prediction (frozen): j4/j16 upper end < 1 at k = 20 (point 0.946), j2 fails (> 1.5x). Why it could beat the current best: a pure set-up saving of 0.12 sqrt(r) (6 %) at k = 20. **Outcome: falsified.** j4 costs 1.220x [1.186, 1.256]; the walk lengthens 34 % and needs 2.93 collisions for two independent rows (dependent re-collisions along the stored path); j2 3.25x as predicted; 16 jumps is not beaten by any tested count (8 jumps ties at k = 20, 1.008 [0.967, 1.051]; 32 jumps ties at k = 24, 1.014 [0.978, 1.051]; 2 and 4 are worse) [pcc:unverified].
2. **Scaling of the target-in-walk arm to k = 28, 32, with 32 and 64 jumps** (ran this session as `curve25519_tw_scaling_20261108`). Mechanism: rho-neg's set-up overhead (0.38 sqrt(r) at k = 20) vanishes with size while twcj's set-up and verification also shrink; and the shorter walk of more jumps (−2.5 % at k = 20) costs nothing once the set-up is ≤ 0.01 sqrt(r). Test: rho-neg, f1-x07, twcj-j16/j32/j64, 8 sessions × 64 rounds at k = 28, 32. Prediction (frozen): twcj/rho-neg point in [1.30, 1.50], j32/j16 in [0.95, 1.00]. **Outcome: 6 of 7 held (P4b failed)** 1.395x [1.345, 1.446] and 1.366x [1.267, 1.473]; j32 0.997 and 0.985 (nulls); the distance is flat in k and sits at the ratio of the two measured walks (1.37; the frozen model predicted 1.38 and 1.375, and the data agree within 1.1 %) rather than the floor ratio 1.5, because rho-neg's search runs 9 to 13 % above 0.886 while twcj's walk runs 3 % above 1.329 [pcc:unverified].
3. **Drop the per-collision recomputation in `TargetWalk`** (needs engine work: an option in `ic_boundary.rs`). Mechanism: each collision is verified by recomputing both entries (0.255 sqrt(r) at k = 20, 12 % of the cost; 0.007 sqrt(r) at k = 32); keys are exact abscissae with a sign bit and the final [d]G = Q check is charged anyway, so a false key collision would only fail that check. Test: `tw=1,cj=1,verify=0` vs the current arm, 8 sessions × 64 rounds at k = 20. Prediction: ratio 0.88 to 0.90 at k = 20 (point 0.88, derived from the phase table as 1 - 0.256 / 2.136 = 0.880) [pcc:unverified], 0.995 at k = 32 [pcc:unverified]; distance to rho 1.24x at k = 20 [pcc:unverified]. Why: it is the largest remaining non-walk term at small k; it does not change the walk or the floor, so it cannot reach parity [pcc:unverified]. Not run: no build (disk), and it is a code change.
4. **Chained set-up for the rho-neg reference** (engine work on `rho.negation`). Mechanism: the `cj` trick applies unchanged to rho's r-adding jumps [a_j]G + [b_j]Q; rho-neg's set-up is 0.376 sqrt(r) at k = 20. Prediction (revised after validator erratum E8): the saving is at most about 0.08 to 0.14 sqrt(r), derived from the tw_jumps set-up GAE (the chained target-in-walk set-up costs 85.8 GAE with 2 jumps and 106.2 with 8, rho-neg uses 8; against rho-neg's whole set-up of 135 GAE, 0.376 sqrt(r), a chained version saves about 49 GAE with 2 jumps and 29 GAE with 8), so rho-neg S goes from 1.53 to about 1.43 to 1.49 at k = 20 and every k = 20 distance grows by 1.05x to 1.09x (the earlier "about 1.25, 1.2x" is withdrawn: it exceeded the whole set-up); no change at k = 32 [pcc:unverified]. Why it matters: it is the honest reference, not an IC gain; it would move the k = 20 "1.4x" figure toward the k = 32 one [pcc:unverified].
5. **Digit-set base in ecbench via `explicit-x` with the `mitm` pair table or the exhaustive grid** (existing tools; the greedy digit sets of L14 at t = 5, 6 are on disk in `curve25519_leads_wave4_20261104`). Mechanism: 2.3 to 2.5x the covered points per column of the best translation frame at equal column count. Test: `explicit-x` (22 columns, 522 covered) + `mitm` vs proj-a (32 columns) at k = 20, 8 sessions. Prediction (derived): trials per relation 251 vs 124, F + 1 = 23 vs 33 rows ⟶ 1.4x proj-a at w = 0, i.e. about 11x rho-neg; it loses to proj-a and is 8x behind the walk arms [pcc:unverified]. Why listed: it is the only algebraic-track base not found run end to end in the specs; it cannot approach rho, so it is ranked below the F = 1 knobs [pcc:unverified].
6. **Odd Kummer base μ_d with a half-gcd decision** (engine work). Mechanism: the D2 norm-resultant decision is Ω(d) per trial; the validator estimated a quasi-linear implementation at "tens" rather than 1,717x the per-point decision. Prediction (derived from the screen): even at 10x the per-point cost the design stays ≥ 10^3x behind at w = 0.1, because coverage d ≈ 2,394 needs 1,208 unknowns [pcc:unverified]. Why listed: it is the last named algebraic family without an engine; the screen says it cannot reach parity [pcc:unverified].
7. **Multi-target amortisation of the F = 1 table (`f1-x07`, not `tw`) against `rho.multitarget_pooled` on its own gauge** (existing tools; outside the single-target question). Mechanism: the F = 1 table is target-independent and reusable; rho's batch grows like sqrt(L). Prediction (derived, Kuhn–Struik): the table arm wins a window of L only if its per-target probe cost (0.58 sqrt(r)) is below rho's marginal cost at that L, which is about 0.886 sqrt(r)/(2 sqrt(L)) for large L, so no window for L ≥ 1 at these constants [pcc:unverified]. Why listed: it is the one setting in which an IC-shaped table beats a rho-shaped walk in the literature; here the constants say no [pcc:unverified].
8. **Linear-algebra variants (`structured-gauss`, `wiedemann`, `lanczos`) on the proj-a or ksum arms.** Not found in specs; prediction: no change in S (LA is < 0.1 % of the cost at F ≤ 32) [pcc:unverified]. Listed only for completeness of the coverage table; not worth a session.

Ideas checked and dropped because they were already tried (with where): more or fewer jumps on the F = 2 walk (round 34, against the fixed-table incumbent only), interleaving (PROPOSALS-R3 P3), stride probes of every kind (rounds 12, 27 to 30), free rows (rounds 13 to 15), the C = 1 boundary (`autolab/c1`), lifts and torsion-shifted targets (T1), coset-pure bases (atlas §5), the Edwards G = 0 anomaly (T6), m = 3 (T3), 2-descent (T2), T_0 frames (T7), the 2-torsion quotient (R4-I2), every structure lever (`structure_vs_rho`), and the BSGS-shaped full-coverage designs (oracle search D1, D4, D5).

4. What the two studies add to the ladder

arm (k)	distance to rho-neg, w = 0	w = 0.1 (pre-2026-11-08 / inserts priced)	w = 1 (pre-2026-11-08 / inserts priced)	source
twcj j16 (20)	1.358 [1.232, 1.497]	1.269 / 1.351	0.804 (artifact) / 1.316	`tw_jumps`
twcj j16 (24)	1.343 [1.219, 1.479]	1.233 / 1.338	0.712 (artifact) / 1.317	`tw_jumps`
twcj j16 (28)	1.395 [1.345, 1.446]	1.275 / 1.397	0.722 (artifact) / 1.407	`tw_scaling`
twcj j16 (32)	1.366 [1.267, 1.473]	1.244 / 1.367	0.689 (artifact) / 1.368	`tw_scaling`
twcj j4 (20)	1.657 [1.522, 1.805]	1.548 / 1.659	0.981 / 1.667	`tw_jumps`

Neither study moves the distance remaining; the current best stays the 16-jump target-in-walk arm at 1.34x to 1.40x rho-neg at w = 0 from k = 20 to 32 [pcc:unverified]. Every number in the table above is an unverified assertion [pcc:unverified]. Under the pre-2026-11-08 reading (inserts unpriced) ecbench's own verdict for rho-neg is "unresolved inside the band" for twcj-j16, j8, j32 (and j64) in every session, for twcj-j4 in 11 of 16 sessions, and "rho-neg ahead" for twcj-j2 (16 of 16) and for f1-x07 (15 of 16 at k = 28, 32) (the w = 1 order flips because the walk arms' per-step table inserts are not priced while rho's canonicalisations are) [pcc:unverified]. With inserts priced, the reading that is now the default, rho-neg is ahead at every weight, by 1.32x to 1.41x; "rho-neg remains ahead at every weight" is true under the priced reading only [pcc:unverified]. The owner ruling of 2026-11-08 (PR #3726) settled how the band prices table inserts; PR #3700 had made it opt-in [pcc:unverified].

The figure `figures/ladder.{svg,png}` plots, for the F = 1 rows, both the pre-2026-11-08 w = 0.1 value (open square: 1.43, 1.31, 1.27) and the inserts-priced value (open diamond: 1.49, 1.39, 1.35); the pre-2026-11-08 square flatters the arm and is keyed as such. Rows 7 to 10 (ksum3, wsum C = 3, wsum2, wsum2g) have no w = 0.1 distance in their sources and carry no w = 0.1 marker [pcc:unverified].

5. Validation

Source: `validator-report.json` in this directory (independent validator, 2026-11-08; status pass_with_errata; not edited). Each erratum and how it was applied [pcc:unverified]:

- **E1** (ladder data): row 1 w01 set to 1.05e6 (1.735e6 / 1.654 = 1.049e6); the w01 of rows 7 to 10 is null (absent) because their sources give only a w = 0 distance and a w = 1 ratio; the figure and this PDF were regenerated from the corrected `data/ladder.json` [pcc:unverified].
- **E2** (count): "5 of 7 held" corrected to 6 of 7 (P4b failed) in sections 0 and 3 [pcc:unverified].
- **E3** (band verdicts): section 4 states the per-arm tally (j16, j8, j32, j64 unresolved in every session; j4 in 11 of 16; j2 and f1-x07 resolved in rho-neg's favour), and the figure plots the inserts-priced F = 1 values as a second, keyed marker [pcc:unverified].
- **E4** (ladder vs floors): "the measured ladder matches" replaced by the two statements that hold and by the statement that boxes and F = 1 distances are against measured rho-neg [pcc:unverified].
- **E5** (classification): the "never tried" rows are split into n/a by `applies` and applicable but not found; prime-kummer is given as run at t = 2 (smoke only, `pkm-smoke-k20`); `walk-prime-fast` is "not found in specs; relabel by help text" [pcc:unverified].
- **E6** (jump-count overclaims): "16 is the optimum" replaced by "not beaten by any tested count"; round 34 listed as tried against the fixed-table incumbent only [pcc:unverified].
- **E7** (scope): "never tried" reworded as "not found in `crypto/research` specs", with the scan (date, signature, size limit) under section 1 [pcc:unverified].
- **E8** (chained set-up prediction): replaced by the supported saving of 0.08 to 0.14 sqrt(r), S about 1.43 to 1.49, distances up 1.05x to 1.09x [pcc:unverified].
- **E9** (tags and numerics): the blanket tag was repeated under tables 2b, 2c and 4 and per-sentence tags were added to section 3 items 3 to 8; "point 0.885" corrected to 0.880 and "0.99 at k = 32" to 0.995 [pcc:unverified].
- **E10** (stale ruling): the accounting ruling is made (owner, 2026-11-08, PR #3726); every "committed band" figure is labelled "the pre-2026-11-08 reading" and both readings are shown [pcc:unverified].
- **E11** (wording and omissions): "only selection" replaced by selection (16x) and trace replay (2.1x); `curve25519_autolab_parity_20261105` and the oracle-search not-preregistered caveat added to section 1 [pcc:unverified].

Disclosures [pcc:unverified]:

- The measuring binary `d23fab99...` named in section 1 was rebuilt on disk at 2026-11-08T21:08:01Z during validation (now `fcb16d9c...`); replays used `fcb16d9c` and reproduced all counts of the two cited studies, so they establish behavioural identity only [pcc:unverified].
- The benchrec records of the two cited studies store 4-decimal values that differ from recomputation by at most 0.07 % (rounding in the producer's pipeline); the raw archives are not uploaded and `session.json` `git_commit` is null [pcc:unverified].
- The validator exported `CHECKPOINT_SKIP` as well as `BENCHREC_SKIP` for its replays; no ledger bypass entry resulted [pcc:unverified].

- The validator traced the headline cell of each row family, not every cell; autolab rounds 1 to 8 beyond the table row, R4-I2 details, the multi-curve study, PROPOSALS-R3 outcomes and the “equal S” of round 5 were not independently traced [pcc:unverified].

6. Files

`REPORT.md`, `REPORT.pdf` (pandoc -t typst + the typst module, `scripts/render_pdf.py`), `data/ladder.json` (every ladder value with its source), `scripts/make_figure.py`, `figures/ladder.{svg,png}`. Studies: `research/curve25519_tw_jumps_20261108/`, `research/curve25519_tw_scaling_20261108/` (protocols frozen at commits `a1ce0db2f9` and `e6a759cc72` before any run; benchmark records `benchmarks/records/20261108T20{45,47,51}Z-curve25519-tw-*.json`, closed).